

PART III

Exercise Training

The study of exercise physiology relies heavily on the understanding of (1) how the body responds during acute bouts of exercise and (2) how it adapts to repeated exercise sessions (i.e., training responses). In the two previous sections of the book, we examined the control and function of skeletal muscle during acute exercise ([part I](#)) and the roles of the cardiovascular and respiratory systems in supporting those functions ([part II](#)). In [part III](#), we examine how these systems adapt when exposed to repeated bouts of exercise (i.e., adaptations to training). [Chapter 9](#), Principles of Exercise Training, lays the groundwork for subsequent chapters by discussing the terminology and training principles used by exercise physiologists. The principles presented in this chapter can be used to optimize the physiological adaptations to a training program. In [chapter 10](#), Adaptations to Resistance Training, we consider the mechanisms through which muscular strength and muscular endurance improve in response to resistance training. Finally, in [chapter 11](#), Adaptations to Aerobic and Anaerobic Training, we discuss the changes in various systems of the body that result from performing regular physical activity involving a wide variety of combinations of exercise intensity and duration. Training adaptations that ultimately lead to improvements in exercise capacity and athletic performance are specific to all aspects of the training to which those physiological systems are exposed.





9

Principles of Exercise Training

In this chapter and in the web study guide

Terminology

Muscular Strength

Muscular Power

Muscular Endurance

Aerobic Power

Anaerobic Power

ACTIVITY 9.1 *Basic Training Principles* reviews the basic training principles and connects them to a real-life situation.

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Principle of Individuality

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AUDIO FOR FIGURE 9.4 explains what happens in the muscle during box jumping.

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ACTIVITY 9.3 *Evaluating an Aerobic Training Program* provides an opportunity to evaluate a basic aerobic power training program.

AUDIO FOR FIGURE 9.6 describes interval training at three different intensities for training each energy system.

In Closing

American Ashton Eaton won the gold medal in the decathlon at the 2012 Olympic Games in London, accumulating 8,869 points over the grueling 2-day competition. At the U.S. Olympic Trials in June of that year, Eaton had broken the 9,000-point barrier as well as the world record, previously held by Roman Šebrle of the Czech Republic, whose mark had stood for 11 years. Decathletes are considered by many to be the ultimate athletes, since they have to compete in events that test their speed, strength, power, agility, and endurance. The decathlon is a 2-day event made up of the 100 m sprint, long jump, shot put, high jump, and 400 m run on the first day, and the 110 m hurdles, discus, pole vault, javelin, and 1,500 m run on the second day. Because training is very specific to the sport or event, intense muscular power training to increase the distance one can heave a 16 lb (~7 kg) shot put does little to improve one's 1,500 m run time. Decathletes spend countless hours training specifically for each of their 10 events, fine-tuning their training techniques to maximize performance in each event.

Previous chapters examining the acute response to exercise covered the body's immediate response to a single exercise bout. We now investigate how the body responds to repeated bouts of exercise performed over a period of time—exercise training. When one performs regular exercise over a period of days, weeks, and months, a variety of physiological adaptations occur. The positive adaptations that accompany proper training principles lead to improvement in both exercise capacity and sport performance. With resistance training, muscles become stronger. With aerobic training, the heart and lungs become more efficient at oxygen delivery, and exercise endurance increases. With high-intensity anaerobic training, the neuromuscular, metabolic, and cardiovascular systems adapt to generate more adenosine triphosphate (ATP) per unit of time, thus increasing muscular endurance and speed of movement over short periods of time. These adaptations are highly specific to the type of training performed. Before examining specific adaptations to training, this chapter first looks at the basic terminology and general principles used in exercise training and then gives an overview of the elements of proper training programs.

Terminology

Before discussing the principles of exercise training, we first define key terms that will be used throughout the rest of this book.

Muscular Strength

Strength is defined as the maximal force that a muscle or muscle group can generate. Someone with a maximal capacity to bench press 100 kg (220 lb) has twice the strength of someone who can bench press 50 kg (110 lb). In this example, strength is defined as the maximal weight the individual can lift with one single effort. This is referred to as **1-repetition maximum (1RM)**. To determine 1RM in the weight room or fitness center, people select a weight that they know they can lift at least one time. After a proper warm-up, they try to execute several repetitions. If they can perform more than one repetition, they add weight and try again to execute several repetitions. This continues until the person is unable to lift the weight more than a single repetition. This last weight that can be lifted only once is the 1RM for that particular exercise. The 1RM is commonly used in the laboratory or weight room as a measure of strength.

Muscular strength can also be accurately measured in the research laboratory through use of specialized equipment that allows quantification of static strength and dynamic strength at various speeds and at various angles in the joint's range of motion (see [figure 9.1](#)). Gains in muscular strength involve changes in both the structure of the muscle and its neural control. These are discussed in [chapter 10](#).



FIGURE 9.1 An isokinetic testing and training device.

Muscular Power

Power is defined as the rate at which work is performed, thus the product of force and velocity. Unlike strength, it has a speed component. Maximal muscular power, generally referred to simply as power, is the explosive aspect of strength, the product of strength and the velocity of movement.

$$\begin{aligned}\text{Power} &= \text{force} \times \text{distance} / \text{time}, \\ &\quad \text{where force} = \text{strength} \\ &\quad \text{and distance} / \text{time} = \text{velocity}\end{aligned}$$

Consider an example. Two individuals can each bench press 200 kg (441 lb), moving the weight the same distance, from where the bar touches the chest to full extension of the arms. But the person who can do it in 1 s has twice the power of the individual who takes 2 s to perform the lift. This is illustrated in [table 9.1](#).

Although absolute strength is an important component of performance, muscular power is the functional application of both strength and speed of movement. It is a key component in almost every sport and competitive activity. In football, for example, an offensive lineman with a bench press 1RM of 200 kg (441 lb) may be unable to control a defensive lineman with a bench press 1RM of only 150 kg (330 lb) if the defensive lineman can move his 1RM at a much faster speed. The offensive lineman is 50 kg (110 lb) stronger, but the defensive lineman's faster speed coupled with adequate strength could give him the performance edge. Although simple field tests are available to estimate power, these tests are generally not very specific to power because their results are affected by other factors. Power can be measured, however, through use of more sophisticated electronic devices, such as the one depicted in [figure 9.1](#).

Throughout this book, the primary concern is with issues of muscular strength, with only brief mention of muscular power. Recall that power has two components: strength and speed. Speed is a more innate quality that changes little with training. Thus, improvements in power follow improvements in strength gained through traditional resistance training programs. However, high power output exercises, such as vertical jump training and some types of resistance training, have been shown to increase power for those specific movements.¹

Muscular Endurance

Many sporting activities depend on the muscles' ability to *repeatedly* develop or sustain submaximal forces or to do both. The capacity to perform repeated muscle contractions, or to sustain a contraction over time, is termed **muscular endurance**. Examples of muscular endurance include performing sit-ups or push-ups or sustaining force in an attempt to pin an opponent in wrestling. Although several valid laboratory techniques are available to directly measure muscular endurance, a simple way to estimate it is to assess the maximum

number of repetitions one can perform at a given percentage of 1RM. For example, a man who has a 1RM for the bench press of 100 kg (220 lb) could evaluate his muscular endurance independent of his muscular strength by measuring how many repetitions he could perform at, for example, 75% of that 1RM (75 kg, or 165 lb). Muscular endurance is increased through gains in muscular strength and through changes in local blood flow and metabolic function. Metabolic and circulatory adaptations that occur with training are discussed in [chapter 11](#).

TABLE 9.1 Strength, Power, and Muscular Endurance of Three Athletes Performing the Bench Press

Component	Athlete A	Athlete B	Athlete C
Strength ^a	100 kg	200 kg	200 kg
Power ^b	100 kg lifted 0.6 m in 0.5 s = 120 kg · m/s = 1,177 J/s or 1,177 W	200 kg lifted 0.6 m in 2.0 s = 60 kg · m/s = 588 J/s or 588 W	200 kg lifted 0.6 m in 1.0 s = 120 kg · m/s = 1,177 J/s or 1,177 W
Muscular endurance ^c	10 repetitions with 75 kg	10 repetitions with 150 kg	5 repetitions with 150 kg

^aStrength was determined by the maximum amount of weight the athlete could bench press just once (i.e., the 1RM).

^bPower was determined as the athlete performed the 1-repetition maximum (1RM) test as explosively as possible. Power was calculated as the product of force (weight lifted) times the distance lifted from the chest to full arm extension (0.6 m, or about 2 ft), divided by the time it took to complete the lift.

^cMuscular endurance was determined by the greatest number of repetitions that could be completed using 75% of the 1RM.

[Table 9.1](#) illustrates the functional differences between strength, power, and muscular endurance in three athletes. The actual values have been exaggerated for the purpose of illustration. From this table we can see that although athlete A has half the strength of athletes B and C, he has twice the power of athlete B and is equal in power to athlete C. Therefore, because of his fast speed of movement, his lack of strength does not seriously limit his power output. Also, for purposes of designing training programs, the analysis of these three athletes indicates that athlete A should focus training on developing strength, without losing speed; athlete B should focus on training explosively to improve speed of movement (although this is unlikely to change much); and athlete C should focus training on developing muscular endurance. These recommendations are made assuming that each athlete needs to optimize performance in each of these three areas.

Aerobic Power

Aerobic power is defined as the rate of energy release by cellular metabolic processes that depend on the continued availability of oxygen. It is synonymous with the terms *aerobic capacity* and *maximal oxygen uptake* ($\dot{V}O_{2\max}$). Maximal aerobic power is the highest oxygen uptake that an individual can obtain during dynamic exercise using large muscle groups for a few minutes. It depends on the maximal capacity for aerobic resynthesis of ATP. In most healthy individuals, maximal aerobic power is limited primarily by the central cardiovascular system and to a lesser extent by respiration and metabolism. The best laboratory test of aerobic power is a graded exercise test to exhaustion during which $\dot{V}O_2$ is measured and $\dot{V}O_{2\max}$ is determined, as discussed in detail in [chapter 5](#). A number of field tests, most often measuring the time needed to walk or run a set distance, or the distance covered in a given time, have been developed to estimate $\dot{V}O_{2\max}$ without the need to actually measure it in the laboratory.

Anaerobic Power

Anaerobic power is defined as the rate of energy release by cellular metabolic processes that function without the involvement of oxygen. Maximal anaerobic power, or *anaerobic capacity*, is defined as the maximal capacity of the anaerobic systems (ATP-phosphocreatine [PCr] system and anaerobic glycolytic system) to produce ATP. Unlike the situation with aerobic power, there is no universally accepted laboratory test to determine anaerobic power. Several tests provide estimates of maximal anaerobic power, including the maximal accumulated oxygen deficit test, the critical power test, and the Wingate anaerobic test.

The commonly used Wingate test involves 30 s of all-out pedaling against a constant resistance on a cycle ergometer. The resistance, or braking force, is determined by the person's weight, sex, age, and level of training. Given a 5 s countdown, subjects begin to pedal as fast as they can, and the resistance is increased instantaneously and held constant for the duration of the test. Peak anaerobic power is determined from the number of revolutions performed in the first 5 s, while anaerobic capacity is measured as the total work performed during the 30 s.

In Review

- Muscular strength refers to the ability of a muscle or muscle group to exert force.
 - Muscular power is the rate of performing work, or the product of force and velocity.
 - Muscular endurance is the capacity to sustain a static contraction or to perform repeated muscle contractions.
 - Maximal aerobic power, or aerobic capacity, is the highest oxygen uptake that an individual can obtain during sustained dynamic exercise using large muscle groups.
 - Maximal anaerobic power, or anaerobic capacity, is defined as the maximal capacity of the anaerobic energy systems to produce ATP.
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General Principles of Training

Chapters 10 and 11 present in detail the specific physiological adaptations that result from resistance training, aerobic training, and anaerobic training. Several principles, however, apply to all forms of exercise training.

Principle of Individuality

Individuals do not all possess the same inherent ability to respond to an acute exercise bout or the same capacity to adapt to exercise training. Heredity plays a major role in determining the body's response to a single bout of exercise, as well as the chronic changes that result from a training program. This is the **principle of individuality**. Except for identical twins, no two people have exactly the same genetic characteristics, so individuals are unlikely to exhibit the same responses. Variations in cellular growth rates, metabolism, cardiovascular and respiratory regulation, and neural and endocrine regulation lead to tremendous individual variation. Such individual variation likely explains why some people show great improvement after participating in a given program ("high responders") whereas others experience little or no change after following the same program ("low responders"). We discuss this phenomenon of high and low responders in more detail in chapter 11. For these reasons, any training program must take into account the specific needs and abilities of the individuals for whom it is designed. Do not expect all

individuals to have exactly the same degree of improvement, even when they train exactly the same.

Principle of Specificity

Training adaptations are highly specific to the type of activity being performed and to the volume and intensity of the exercise. To improve muscular power, for example, a shot-putter would not emphasize distance running or slow, low-intensity resistance training. The shot-putter needs to develop explosive power. Similarly, the marathon runner would not focus on sprint training. This is likely the reason that athletes who train for strength and power, such as weightlifters, often have great strength but don't have highly developed aerobic endurance when compared to untrained people. According to the **principle of specificity**, exercise adaptations are specific to the mode, intensity, and duration of training, and the training program must stress the physiological systems that are critical for optimal performance in a given sport in order to achieve specific training adaptations and goals.

Principle of Reversibility

Resistance training improves muscle strength and the capacity to resist fatigue. Likewise, endurance training improves the ability to perform aerobic exercise at higher intensities and for longer periods. But if training is decreased or stopped (detraining), the physiological adaptations that caused those improvements in performance will be reversed. Any gains achieved with training will eventually be lost. The **principle of reversibility** lends scientific support to the saying "Use it or lose it." All effective training programs must include a maintenance plan that sustains the physiological adaptations gained by training. In [chapter 14](#), we examine specific physiological changes that occur when the training stimulus stops.

RESEARCH PERSPECTIVE 9.1

Can Aerobic Exercise Increase Muscle Size?

The principle of specificity states that training adaptations are highly specific to the type of training performed. Aerobic exercise training is associated with improvements in aerobic capacity and cardiorespiratory function. However, there is debate among exercise physiologists about the impact of aerobic

exercise training on skeletal muscle mass. Historically, it has been assumed that aerobic exercise has little effect on skeletal muscle hypertrophy. With the development of high-resolution imaging techniques, accumulating evidence now suggests that aerobic exercise training can improve muscle mass in sedentary individuals across the life span. The first of these studies established that 6 months of walking or running training could increase the cross-sectional area of the thigh by 9% in older men.¹³ In that study, while the older men experienced a robust increase in muscle size, a group of young men did not show any changes in muscle size with the training. (This result may have been because the younger men attended fewer exercise sessions than the older men did throughout the study. The effectiveness of aerobic exercising training to induce skeletal muscle hypertrophy likely depends on obtaining sufficient exercise intensities, duration, and frequency to accumulate a large number of muscle contractions at this lower load.) Studies that have compared aerobic training with resistance training have found that, on average, both modalities increase muscle size by approximately the same percentage (~7%-9%) from baseline.

Aside from just increasing whole muscle size, aerobic training increased slow- and fast-twitch myofiber cross-sectional area of the exercised muscle in the majority of studies. Similarly, studies of the metabolic turnover of muscle proteins showed that aerobic exercise acutely and chronically stimulated skeletal muscle protein synthesis, resulting in a positive muscle protein balance and increased myofiber size, even in older men and women who might otherwise have age-related anabolic impairments. Despite a lack of standard methodology to measure muscle protein breakdown, most studies examining muscle protein breakdown and aerobic training agree that aerobic training results in reduced catabolic factors, leading to skeletal muscle hypertrophy.

Overall, the existing research indicates that aerobic exercise training can produce skeletal muscle hypertrophy.¹¹ Aerobic exercise-induced changes in the molecular regulation and protein metabolism of skeletal muscle increase both individual myofiber and whole muscle size in sedentary individuals. These data show that aerobic exercise should be acknowledged for its ability to increase skeletal muscle mass and considered an effective countermeasure for age-related muscle atrophy.

Principle of Progressive Overload

Two important concepts, overload and progressive training, form the foundation of all training programs. According to the **principle of progressive overload**, systematically increasing the demands on the body is necessary for continued improvement. For example, when undergoing a strength training program, in order to gain strength the

muscles must be overloaded, which means they must be loaded beyond the point to which they are normally loaded. Progressive resistance training implies that as the muscles become stronger, either increased resistance or increased repetitions or both are required to stimulate further strength increases.

As an example, consider a young woman who can perform only 10 repetitions of a bench press before reaching fatigue, using 30 kg (66 lb) of weight. With a week or two of resistance training, she should be able to increase to 14 or 15 repetitions with the same weight. She then adds 2.3 kg (5 lb) to the bar, and her repetitions decrease to 8 or 10. As she continues to train, the repetitions continue to increase, and within another week or two, she is ready to add another 2.3 kg of weight. Thus, improvement depends on a progressive increase in the amount of weight lifted. In a similar way, training volume (intensity and duration) must be increased progressively with anaerobic and aerobic training for further improvements to occur.

Principle of Variation

The **principle of variation**, also called the **principle of periodization**, first proposed in the 1960s, has become very popular in the area of resistance training. Periodization is the systematic process of changing one or more variables in the training program—mode, volume, or intensity—over time to allow for the training stimulus to remain challenging and effective.¹ Training intensity and volume of training are the most commonly manipulated aspects of training to achieve peak levels of fitness for competition. Classical periodization involves high initial training volume with low intensity; then, as training progresses, volume decreases and intensity gradually increases. Undulating periodization uses more frequent variation within a training cycle.

For sport-specific training, the volume and intensity of training are varied over a macrocycle, which is generally up to a year of training. A macrocycle is composed of two or more mesocycles that are dictated by the dates of major competitions. Each mesocycle is subdivided into periods of preparation, competition, and transition. This principle is discussed in greater detail in [chapter 14](#).

In Review

- According to the principle of individuality, each person responds uniquely to training, and training programs must be designed to allow for individual variation.
 - According to the principle of specificity, to maximize benefits, training must be specifically matched to the type of activity or sport the person engages in. An athlete involved in a sport that requires tremendous strength, such as weightlifting, would not expect great strength gains from endurance running.
 - According to the principle of reversibility, training benefits are lost if training is either discontinued or reduced abruptly. To avoid this, all training programs must include a maintenance program.
 - According to the principle of progressive overload, as the body adapts to training at a given volume and intensity, the stress placed on the body must be increased progressively for the training stimulus to remain effective in producing further improvements.
 - According to the principle of variation (or periodization), one or more aspects of the training program should be altered over time to maximize effectiveness of training. The systematic variation of volume and intensity is most effective for long-term progression.
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Resistance Training Programs

Over the past 75 years, research has provided a substantial knowledge base concerning resistance training and its application to health and sport. The health aspects of resistance training are discussed in [chapter 20](#). This section concerns primarily the use of resistance training for sport.

Recommendations for Resistance Training Programs

Resistance training programs can be designed and prescribed in terms of

- the exercises that will be performed;
- the order in which they will be performed;
- the number of sets for each exercise;
- the rest periods between sets and between exercises; and
- the amount of resistance, the number of repetitions, and the velocity of movement to be used.

In 2009, the American College of Sports Medicine (ACSM) revised its position stand on progressive resistance training for healthy adults ([table 9.2](#)).¹ Previous statements specified, for all adults, a minimum of one set of 8 to 12 reps for each of 8 to 10 different exercises that together involve all of the major muscle groups. The new position stand recommends resistance training models specific to desired outcomes, that is, improvements in strength, muscle **hypertrophy**, power, local muscular endurance, or gross motor performance.

Resistance programs aimed at improving strength should involve repetitions with both concentric (muscle shortening) and eccentric (muscle lengthening) actions. Isometric contractions play a beneficial, but secondary, role and may be included as well. Concentric strength improvement is greatest when eccentric exercises are included, and eccentric training has been shown to produce specific benefits for those action-specific movements. Large muscle groups should be stressed before smaller groups, multiple-joint exercises before single-joint exercises, and higher-intensity efforts before those of lower intensity. [Table 9.2](#) provides a summary of the ACSM recommendations on loading, volume (sets and reps), velocity of movements, and frequency of training.

TABLE 9.2 American College of Sports Medicine's Recommendations for Resistance Training Programs^a

Primary goal of resistance training program	Training level	Loading	Volume	Velocity	Frequency (times per week)
Strength development	Novice	60%-70% 1RM	1-3 sets, 8-12 reps	Slow, moderate	2-3
	Intermediate	70%-80% 1RM	Multiple sets, 6-12 reps	Moderate	3-4
	Advanced	80%-100% 1RM	Multiple sets, 1-12 reps	Unintentionally slow to fast	4-6
Muscle hypertrophy	Novice	70%-85% 1RM	1-3 sets, 8-12 reps	Slow, moderate	2-3
	Intermediate	70%-85% 1RM	1-3 sets, 6-12 reps	Slow, moderate	4
	Advanced	70%-100% 1RM; emphasis on 70-85%	3-6 sets, 1-12 reps	Slow, moderate, fast	4-6
Development of muscle power	Novice	0%-60% 1RM—lower body; 30%-60% 1RM—upper body	1-3 sets, 3-6 reps	Moderate	2-3
	Intermediate	0%-60% 1RM—lower body; 30%-60% 1RM—upper body	1-3 sets, 3-6 reps	Fast	3-4
	Advanced	85%-100% 1RM	3-6 sets, 1-6 reps, various strategies ^b	Fast	4-5
Increased local muscular endurance	Novice	Light	1-3 sets, 10-15 reps	Slow—moderate reps Moderate—high reps	2-3
	Intermediate	Light	1-3 sets, 10-15 reps	Slow—moderate reps Moderate—high reps	3-4
	Advanced	30%-80% 1RM	Various strategies, ^b 10-25 reps or more	Slow—moderate reps Moderate—high reps	4-6

^aThese recommendations also include the type of muscle action (eccentric and concentric), single-joint versus multiple-joint exercises, order or sequence of exercises, and the rest intervals. See text for further information.

^bPeriodized—see text for explanation of periodization.

Adapted from information in ACSM (2009).

It is recommended that rest periods of 2 to 3 min or more be used between heavy loads for novice and intermediate lifters; for advanced lifters, 1 or 2 min may suffice. Once an individual can perform the current workload at or above the desired number of reps for two consecutive training sessions, a 2% to 10% increase in load should be applied. While both machine-based exercises and free weights can be used for novice and intermediate lifters, for advanced lifters, the emphasis should be placed on free weights.

When muscle hypertrophy (in bodybuilders, for example) or development of muscular power is the goal, recommendations for sequencing, rest periods, and so on are the same as those for

strength development. However, as shown in [table 9.2](#), other aspects of the program differ.

Types of Resistance Training

Resistance training can use static contractions, dynamic contractions, or both. Dynamic contractions can include concentric or eccentric contractions, or both. Typical resistance training can be performed using free weights, variable-resistance devices, isokinetic devices, and plyometrics.

Static-Contraction Resistance Training

Static-contraction resistance training, also called **isometric training**, gained great popularity in the mid-1950s as a result of research by several German scientists. These studies indicated that static resistance training caused tremendous strength gains and that those gains exceeded the gains resulting from dynamic-contraction procedures. Subsequent studies were unable to reproduce the original studies' results, and training programs based heavily on isometric contractions have generally fallen out of favor. However, static contractions remain an important form of training, particularly for core stabilization (discussed later in the chapter) and for enhancing grip strength.¹ Additionally, in postsurgical rehabilitation when a limb is immobilized and thus incapable of dynamic contractions, static contractions facilitate recovery and reduce muscle atrophy and strength loss.

Free Weights Versus Machines

With **free weights**, such as barbells and dumbbells, the resistance or weight lifted remains constant throughout the dynamic range of movement. If a 50 kg (110 lb) weight is lifted, it will always weigh 50 kg. In contrast, a variable-resistance contraction involves varying the resistance to try to match it to the strength curve. [Figure 9.2](#) illustrates how strength varies throughout the range of motion in a two-arm curl. Maximal strength production by the elbow flexors occurs at approximately 100° in the range of movement. These muscles are weakest at 60° (elbows fully flexed) and at 180° (elbows fully extended). In these positions, one is able to generate only 67%

and 71%, respectively, of the maximal force-producing capabilities at the optimal angle of 100° .

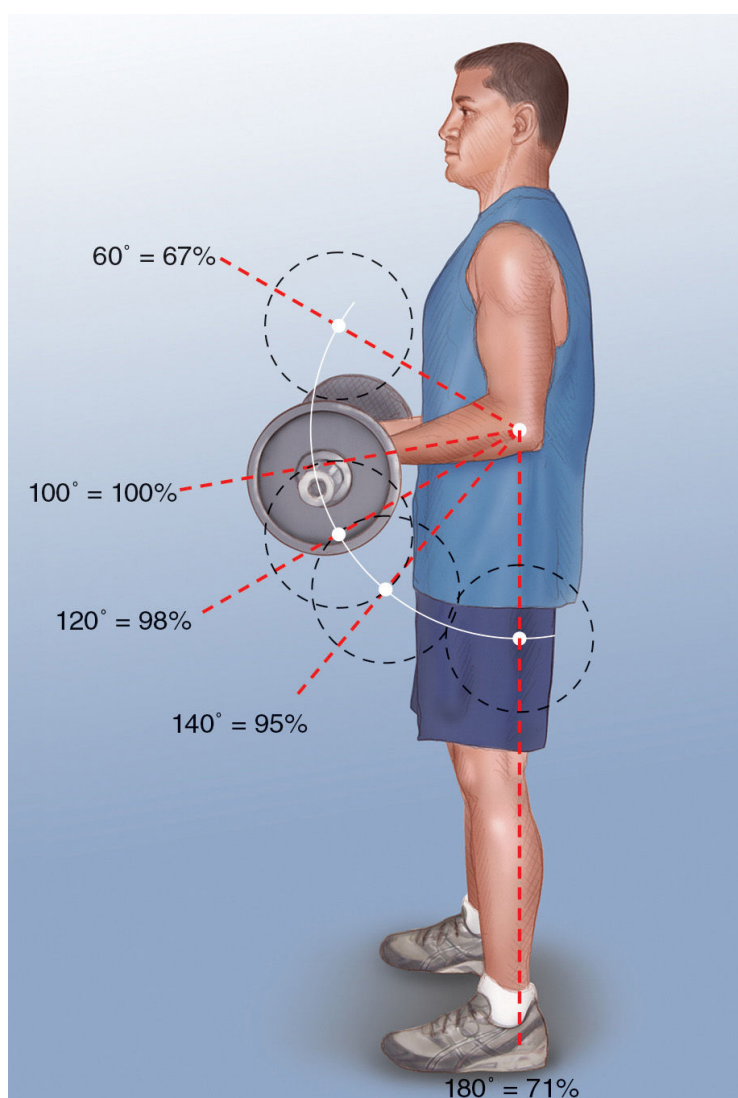


FIGURE 9.2 The variation in strength relative to the angle of the elbow during the two-arm curl. Strength is optimized at an angle of 100° . The maximal force-development capacity of the muscle group at a given angle is given as a percentage of the capacity at the optimal angle of 100° .

When one is using free weights, the range of motion is less restricted than with machines, and the resistance or weight used to train the muscle is limited by the weakest point in that range of motion. If the person in [figure 9.2](#) had the capacity to lift only 45 kg (100 lb) at the optimal angle of 100° , then he would be able to lift only 32 kg (71 lb) at the fully extended position of 180° . Therefore, if he is starting with a barbell loaded with 32 kg, he can just barely move it

from the fully extended position to start his lift. However, by the time he gets to an angle of 100° in his full range of motion, he is lifting only 70% of what he could maximally lift at that angle. Thus, free weights maximally tax the weakest points in the range of motion and provide moderate resistance at the midrange (90° - 140°). Individuals performing the two-arm curl tend to greatly reduce their range of motion as they start to fatigue (referred to as “cheating”). They are simply trying to stay out of the weakest portion of their range of motion. The bottom line is that with free weights, the maximum weight one can lift is limited by the weakest portion in the range of motion, which means that the strongest position in the range of motion is never maximally taxed! However, free weights do offer some distinct advantages, especially for the expert lifter.

Starting in the 1970s, a number of resistance training machines or devices were introduced that used stacked weights, variable-resistance, and isokinetic techniques. Variable-resistance machines use cams, pulleys, and levers to vary the weight throughout the range of movement. Such machines have been regarded as safer; they are easy to use and allow performance of some exercises that are difficult to do with free weights. Machines help stabilize the body, especially for novice lifters, and limit the muscle action to that desired without extraneous muscle groups firing.

On the other hand, free weights offer some advantages that resistance machines do not provide. The lifter must control the weight being lifted. A lifter must recruit more motor units—not only in the muscles being trained but also in supporting muscles—to gain control of the bar, stabilize the weight lifted, and maintain body balance. The lifter must both balance and stabilize the weight. In that regard, when an athlete is training for a sport such as football, the experience with free weights more closely resembles actions associated with actual sport competition. Also, because free weights do not limit the range of motion of a particular exercise, optimal training specificity can be achieved. Whereas a bicep curl on a machine may be done only in the vertical plane, an athlete using free weights can perform the curl in any plane, choosing one, for example, that reflects a sport-specific motion. And finally, data show that if significant strength gains are to

be achieved over a shortened training period, free weights may provide greater strength gains than many types of weight machines.

Both machine-based resistance programs and free-weight training programs result in measurable gains in strength, hypertrophy, and power. Free-weight programs result in greater improvements in free-weight tests, and machine training results in greater gains in machine-based tests. The choice to use weight machines versus free weights depends on the experience of the lifter and the desired outcomes.

Eccentric Training

Another form of dynamic-contraction resistance training, called **eccentric training**, emphasizes the eccentric phase. With eccentric contractions, the muscle's ability to resist force is considerably greater than with concentric contractions (see [chapter 1](#)). Subjecting the muscle to this greater training stimulus theoretically produces greater strength gains. A number of studies have shown the importance of including the eccentric phase of muscle contraction along with the concentric phase to maximize gains in muscle strength and size. Further, eccentric contraction is important to stimulate muscle hypertrophy, as discussed in the next chapter.

Variable-Resistance Training

With a variable-resistance device, the resistance is decreased at the weakest points in the range of movement and increased at the strongest points. **Variable-resistance training** is the basis for several popular resistance training machines. The underlying theory is that the muscle can be more fully trained if it is forced to act at higher constant percentages of its capacity throughout each point in its range of movement. [Figure 9.3](#) illustrates a variable-resistance device in which a cam alters the resistance through the range of motion. As noted earlier, there are advantages and disadvantages to training using such machines.



FIGURE 9.3 A variable-resistance training device that uses a cam to alter the resistance through the range of motion.

Isokinetic Training

Isokinetic training is conducted with equipment that keeps movement speed constant. Whether one applies very light force or an all-out maximal muscle contraction, the speed of movement does not vary. Using electronics, air, or hydraulics, the device can be preset to control the speed of movement (angular velocity) from $0^\circ/\text{s}$ (static contraction) to $300^\circ/\text{s}$ or higher. An isokinetic device is illustrated in [figure 9.1](#). Theoretically, if properly motivated, the individual can contract the muscles at maximal force at all points in the range of motion.

Plyometrics

Plyometrics, or stretch–shortening cycle exercise, became popular during the late 1970s and early 1980s, primarily for improving jumping ability. As an example, to develop knee extensor muscle strength and power, a person goes from standing upright to a deep squat position (eccentric contraction) and then jumps up onto a box (concentric contraction), landing in a squat position on top of the box. The person then jumps off the box onto the ground, landing in a squat position, and repeats the sequence with the next box (see [figure 9.4](#)).

Proposed to bridge the gap between speed and strength training, plyometrics uses the stretch reflex to facilitate recruitment of motor units. It also stores energy in the elastic and contractile components of muscle during the eccentric contraction (stretch) that can be recovered during the concentric contraction.

Electrical Stimulation

One can stimulate a muscle by passing an electric current directly across it or its motor nerve. This technique, called **electrical stimulation**, has proven effective in clinical settings to reduce the loss of strength and muscle size during periods of immobilization and to restore strength and size during rehabilitation. Electrical stimulation training also has been used experimentally in healthy subjects (including athletes). Athletes have used this technique to supplement their regular training programs, but no evidence shows any additional gains in strength, power, or performance from this supplementation.

Core Training

In recent years, a significant emphasis has been placed on core stability and strengthening exercises. While there are varying opinions on what anatomical features constitute the core, the general consensus is that the core is the group of trunk muscles that surround the spine and abdominal viscera and include the abdominal, gluteal, hip girdle, paraspinal, and other accessory muscles.

Initially, this type of core-specific exercise training was explored in rehabilitation settings, specifically for the treatment of lower back pain, but its benefits have also been recognized in sport performance. Theoretically, greater core stability could benefit sport performance by providing a foundation for greater force production and force transfer

to the extremities. For example, having the core stabilized and engaged in the simple action of throwing a ball allows for greater biomechanical efficiency in the limb transmitting the force to throw the ball and for the activation of stabilizing muscles in the contralateral arm. The principle of core stabilization promotes proximal stability for distal mobility.



FIGURE 9.4 Plyometric box jumping (see the text for a detailed explanation).



There has been little definitive research on the benefits of core stability and core strengthening for athletic performance. One reason is that there are no standardized tests for evaluating core strength and stability. Further, the studies that have been done have been mainly with injured populations and not specific to athletic performance. However, the limited research does show that this type of training decreases the likelihood of injury, especially in the lower back and the lower extremities, during sport performance. The physiological explanation for this finding is that core stability training increases the sensitivity of the muscle spindles, thereby permitting a

greater state of readiness for loading joints during movement¹⁵ and protecting the body from injury.

The many different types of core stability and strengthening training include balance and instability resistance (e.g., physioball). It is thought that because the core is composed mainly of type I muscle fibers, the core musculature may respond well to multiple sets of exercises with high repetitions.⁴ Yoga, Pilates, tai chi, and the physioball are commonly incorporated into athletes' training programs to promote core stability and strength. Further research is needed to determine the benefits of core training and the underlying mechanisms.

In Review

- Low-repetition, high-resistance training enhances strength development, whereas high-repetition, low-intensity training optimizes the development of muscular endurance.
- Variation (or periodization), through which various aspects of the training program are altered, is important to optimize results and prevent overtraining or burnout.
- Resistance programs aimed at improving strength should involve repetitions with both concentric (muscle shortening) and eccentric (muscle lengthening) actions. Isometric contractions play a beneficial, but secondary, role and may be included as well.
- Large muscle groups should be stressed before smaller groups, multiple-joint exercises before single-joint exercises, and higher-intensity efforts before those of lower intensity.
- Rest periods of 2 to 3 min or more should be incorporated between heavy loads for novice and intermediate lifters; for advanced lifters, 1 to 2 min may suffice.
- The ability of a muscle or muscle group to generate force varies throughout the full range of movement.
- While both machine-based exercises and free weights can be used for novice and intermediate lifters, for advanced lifters, the emphasis should be placed on free weights.
- When neutral testing devices are used, strength gains from free-weight programs and machine-based programs are similar.
- Electrical stimulation can be successfully used in rehabilitating athletes but has no additional benefits when used to supplement resistance training in healthy

athletes.

- Exercises aimed at improving core stability may benefit sport performance by providing a foundation for greater force production and force transfer to the extremities while stabilizing other parts of the body. However, direct evidence of such a benefit is lacking.
-

Anaerobic and Aerobic Power Training Programs

Anaerobic and aerobic power training programs, while quite different at the extremes (e.g., training for the 100 m dash versus the 42.2 km [26.2 mi] full marathon), are designed along a continuum. [Table 9.3](#) illustrates how training requirements vary in competitive running events as one goes from short sprints to long distances. With this table serving as an example that can be applied to all sports, the primary emphasis for the short sprints is on training the ATP-PCr system. For longer sprints and middle distances, the primary emphasis is on the glycolytic system, and for the longer distances, the primary emphasis is on the oxidative system. Anaerobic power is represented by the ATP-PCr and anaerobic glycolytic systems, while aerobic power is represented by the oxidative system. Note, however, that even at the extremes, more than one energy system must be trained.

Different types of training programs can be used to meet the specific training requirements of each event, such as in running and swimming, and each sport. This section describes some of the more popular types of training programs and how they are used to improve the specific energy systems.

Group Exercise Training

The first description of group fitness can be found in the 1968 book *Aerobics* by Dr. Kenneth Cooper. His mission was to encourage people to exercise with the goal of disease prevention rather than disease treatment. One suggested method was a new form of exercise that utilized dance movements, primarily from hip-hop and jazz, choreographed with music and led by an instructor. Currently, group fitness options focus on varying types of cardiovascular, strength, and flexibility training. For instance, cardio classes include

mixed martial arts, plyometric training, indoor cycling, and aquatic activities. Multiple strength training formats exist that range from high-repetition barbell classes to boot camps with more traditional powerlifting techniques to core-based functional actions. Finally, flexibility is the emphasis in a range of yoga disciplines as well as in fall or injury prevention sessions.

TABLE 9.3 Percentage of Emphasis on the Three Metabolic Energy Systems in Training for Various Running Events

Running event	Anaerobic speed (ATP-PCr system)	Anaerobic endurance (anaerobic glycolytic system)	Aerobic endurance (oxidative system)
100 m (109 yd)	95	3	2
200 m (218 yd)	95	2	3
400 m (436 yd)	80	15	5
800 m (872 yd)	30	65	5
1,500 m (0.93 mi)	20	55	25
3,000 m (1.86 mi)	20	40	40
5,000 m (3.10 mi)	10	20	70
10,000 m (6.20 mi)	5	15	80
Marathon (42.20 km; 26.20 mi)	5	5	90

Adapted from F. Wilt, Training for Competitive Running, in *Exercise Physiology*, edited by H.B. Falls (Amsterdam, Netherlands: Elsevier, 1968).

Group fitness can provide the equivalent health benefits of independent exercise, increasing oxygen consumption, high-density lipoprotein (HDL), and lean muscle mass while decreasing fasting blood glucose, low-density lipoprotein (LDL), triglycerides, and fat mass. These positive physiological results occur in parallel to the improvement of many psychological variables such as satisfaction, enjoyment, challenge, and motivation. Because of these health benefits and the prevalence of many class styles, durations, and intensities, group fitness can be an ideal recommendation for all ages and abilities.

Interval Training

Interval training consists of repeated bouts of high- to moderate-intensity exercise interspersed with periods of rest or reduced-intensity exercise. Research has shown that athletes can perform a considerably greater total volume of exercise by breaking the overall exercise period into shorter, more intense bouts, with rest or active recovery intervals inserted between the intense bouts.

The vocabulary used to describe an interval training program is similar to that used in resistance training and includes the terms *sets*,

repetitions, training time, training distance and frequency, exercise interval, and rest or active recovery interval. Interval training is frequently prescribed in these terms, as illustrated in the following example for a middle-distance runner:

- Set 1: 6 × 400 m at 75 s (90 s slow jog)
- Set 2: 6 × 800 m at 180 s (200 s jog-walk)

For the first set, the athlete would run six repetitions of 400 m each, completing the exercise interval in 75 s and recovering for 90 s between exercise intervals with slow jogging. The second set consists of running six repetitions of 800 m each, completing the exercise interval in 180 s, and recovering for 200 s with walking-jogging.

While interval training is traditionally associated with track, cross country running, and swimming, it is appropriate for all sports and activities. One can adapt interval training procedures for each sport or event by first selecting the form or mode of training and then manipulating the following primary variables to fit the sport and athlete:

- Rate of the exercise interval
- Distance of the exercise interval
- Number of repetitions and sets during each training session
- Duration of the rest or active recovery interval
- Type of activity during the active recovery interval
- Frequency of training per week

Exercise Interval Intensity

One can determine the intensity of the exercise interval either by establishing a specific duration for a set distance, as illustrated in our previous example for set 1 (i.e., 75 s for 400 m), or by using a fixed percentage of the athlete's maximal heart rate (HR_{max}). Setting a specific duration is more practical, particularly for short sprints. One typically determines this by using the athlete's best time for the set distance and then adjusting the duration according to the relative intensity that the athlete wants to achieve, with 100% equal to the athlete's best time. As an example, to develop the ATP-PCr system, the intensity should be near maximal (e.g., 90%-98%); to develop the

anaerobic glycolytic system, it should be high (e.g., 80%-95%); and to develop the aerobic system, it should be moderate to high (e.g., 75%-85%). These estimated percentages are only approximations and are dependent on the athlete's genetic potential and fitness level, duration of the interval (e.g., 10 s versus 10 min), number of repetitions and sets, and duration of the active recovery interval.

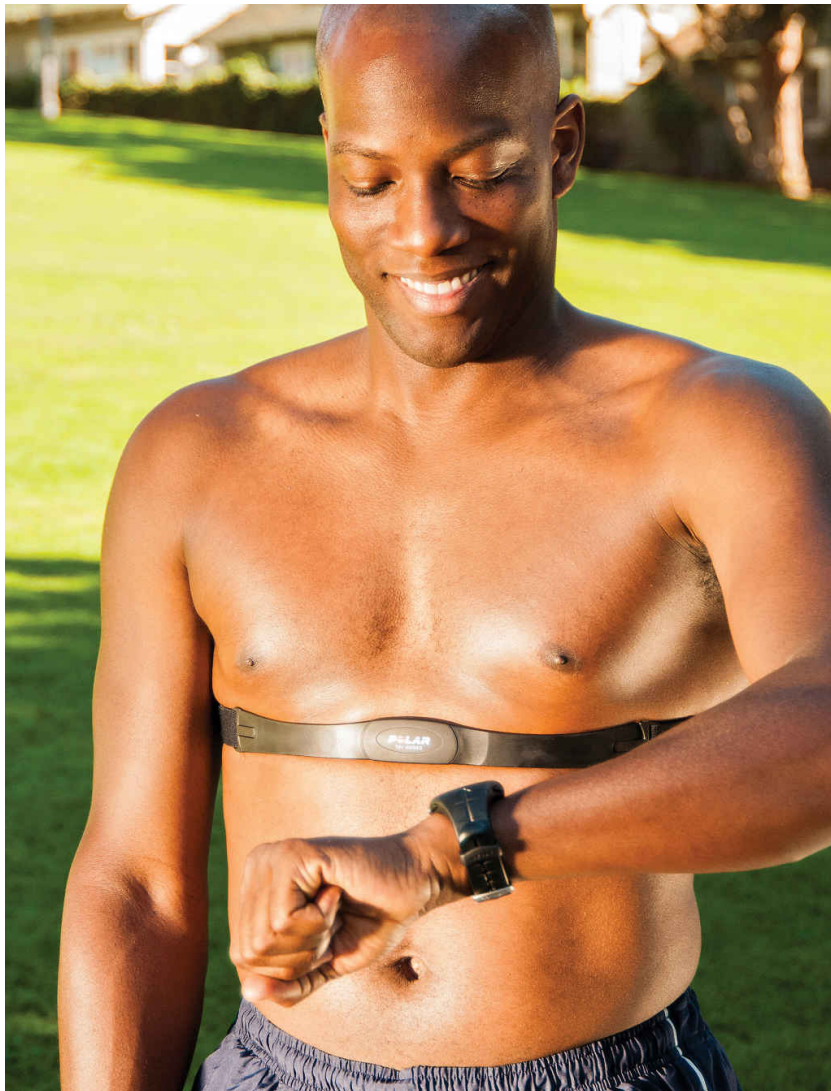


FIGURE 9.5 A runner outfitted with a heart rate monitor. The receiving unit, attached to the chest strap, picks up and transmits electrical impulses from the heart to the digital monitor and memory device worn on the wrist. After the workout, the contents of the memory device can be downloaded to a computer.

Using a fixed percentage of the athlete's $HR_{\max}$ might provide a better index of the physiological stress experienced by the athlete. Heart rate monitors are now readily available and relatively

inexpensive (see [figure 9.5](#)). HR_{max} can be determined during a maximal exercise test in the laboratory as described in [chapter 8](#) or during an all-out run on the track using the heart rate monitor. Training the ATP-PCr system requires training at very high percentages of the athlete's HR_{max} (e.g., 90%-100%), as does training to develop the anaerobic glycolytic system (e.g., 85%-100% HR_{max}). To develop the aerobic system, the intensity should be moderate to high (e.g., 70%-90% HR_{max}).

[Figure 9.6](#) illustrates changes in blood lactate concentration in a runner using interval training at three different intensities corresponding to those intensities needed to train the ATP-PCr system, the glycolytic system, and the oxidative system. The runner performed a single set consisting of five repetitions at each intensity on different days, and the lactate concentrations were obtained from a blood sample taken after the last repetition of each intensity. Monitoring blood lactate concentrations can verify the energy system that is primarily being trained.

Distance of the Exercise Interval

The distance of the exercise interval is determined by the requirements of the event, sport, or activity. Athletes who run or sprint short distances, such as track sprinters, basketball players, and soccer players, use short intervals of 30 to 200 m (33-219 yd), although a 200 m sprinter frequently runs over distances of 300 to 400 m (328-437 yd). A 1,500 m runner may run intervals as short as 200 m to increase speed, but most of this athlete's training would be at distances of 400 to 1,500 m (437-1,640 yd), or even longer distances, to increase endurance and decrease fatigue or exhaustion in the race.

Number of Repetitions and Sets During Each Training Session

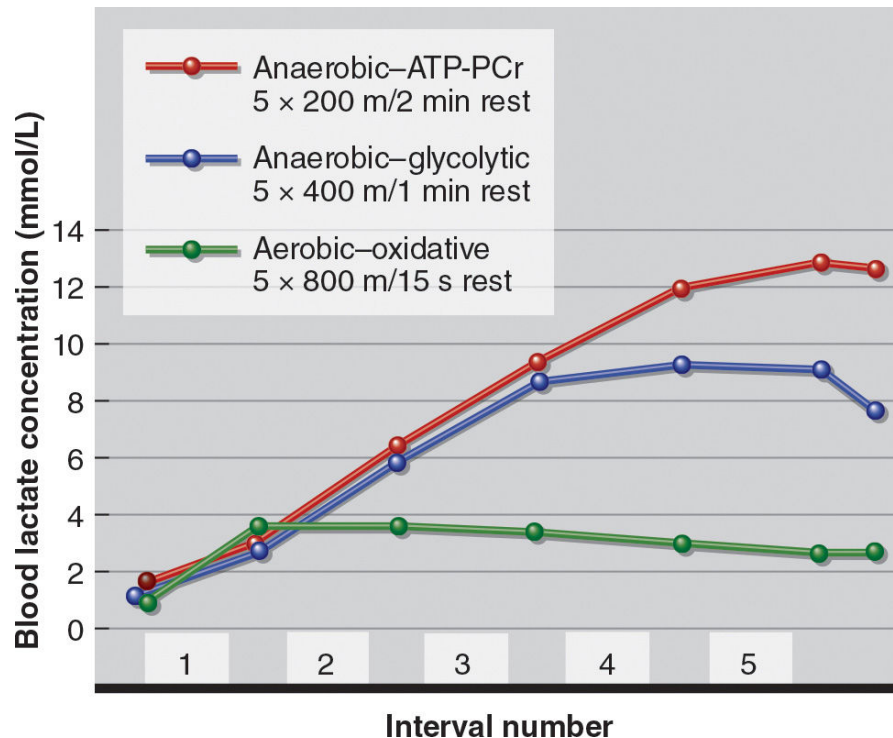


FIGURE 9.6 Blood lactate concentrations in a single runner following a set of five repetitions of interval training at three different paces, each on different days, corresponding to the appropriate pace for training each energy system.



The number of repetitions and sets is also largely determined by the needs of the sport, event, or activity. Generally, the shorter and more intense the interval, the greater the number of repetitions and sets should be. As the training interval is lengthened in both distance and duration, the number of repetitions and sets is correspondingly reduced.

Duration of the Rest or Active Recovery Interval

The duration of the rest or active recovery interval depends on how rapidly the athlete recovers from the exercise interval. The extent of recovery is best determined by the reduction of the athlete's heart rate to a predetermined level during the rest or active recovery period. For younger athletes (30 years of age or younger), heart rate

is generally allowed to drop to between 130 and 150 beats/min before the next exercise interval begins. For those over 30 years, since $HR_{\max}$ decreases ~ 1 beat/min per year, we subtract the difference between the athlete's age and 30 years from both 130 and 150. So, for a 45-year-old, we would subtract 15 beats/min to obtain the athlete's recovery range of 115 to 135 beats/min. The recovery interval between sets can be established in a similar manner, but generally the heart rate should be below 120 beats/min.



Type of Activity During the Active Recovery Interval

The type of activity performed during the active recovery interval for land-based training can vary from slow walking to rapid walking or

jogging. In the pool, slow swimming using alternative strokes or the primary stroke is appropriate. In some cases, usually in the pool, total rest can be used. Generally, the more intense the exercise interval, the lighter or less intense the activity performed in the recovery interval. As athletes become better conditioned, they are able to increase the intensity of the exercise interval, decrease the duration of the rest interval, or both.

Frequency of Training per Week

The frequency of training depends largely on the purpose of the interval training. A world-class sprinter or middle-distance runner typically works out 5 to 7 days a week, although not every workout will include interval training. Swimmers use interval training almost exclusively. Team sport athletes can benefit from 2 to 4 days of interval training per week when interval training is used only as a supplement to a general conditioning program.

Continuous Training

Continuous training involves continuous activity without rest intervals. This can vary from **long, slow distance (LSD) training** to high-intensity endurance training. Continuous training is structured primarily to affect the oxidative and glycolytic energy systems. High-intensity continuous activity is usually performed at intensities representing 85% to 95% of the athlete's HR_{max} . For swimmers and track and cross country athletes, this could be above, at, or near race pace. This pace would likely match or exceed the pace associated with the athlete's lactate threshold. Scientific evidence has clearly demonstrated that marathon runners typically race at, or very close to, their lactate threshold.

Long, slow distance (LSD) training became extremely popular in the 1960s. With this form of training, introduced in the 1920s by Dr. Ernst van Aaken, a German physician and coach, the athlete typically trains at relatively low intensities, between 60% and 80% of HR_{max} , which is approximately the equivalent of 50% to 75% of $\dot{V}O_{2max}$. Distance, rather than speed, is the main objective. Distance runners may train 15 to 30 mi (24-48 km) per day using LSD techniques, with weekly distances of 100 to 200 mi (161-322 km). The pace of the run is substantially slower than the runner's maximal pace. While less

stressful to the cardiovascular and respiratory systems, extreme distances can result in overuse injuries and general breakdown of muscles and joints. Further, the serious runner needs to train at or near race pace on a regular basis to develop leg speed and strength. Thus, most runners vary their workout from one day to the next, from week to week, and from month to month.

Long, slow distance training is probably the most popular and safest form of aerobic endurance conditioning for the nonathlete who just wants to get into shape and stay in shape for health-related purposes. More vigorous or burst types of activity generally are not encouraged in older, sedentary people. Long, slow distance is also a good training program for athletes in team sports for maintaining aerobic endurance during the season as well as the off-season.

Fartlek training, or speed play, is another form of continuous exercise that has some components of interval training. This form of training was developed in Sweden in the 1930s and is used primarily by distance runners. The athlete varies the pace from high speed to jogging speed at his or her discretion. This is a free form of training in which fun is the primary goal, and distance and time are not even considered. Fartlek training is normally performed in the countryside where there are hills of various inclines. Many coaches have used Fartlek training to supplement either high-intensity continuous training or interval training, since it provides variety to the normal training routine.

Interval-Circuit Training

Introduced in the Scandinavian countries in the 1960s and 1970s, **interval-circuit training** combines interval and circuit training into one workout. The circuit may be 3,000 to 10,000 m in length, with stations every 400 to 1,600 m (437-1,750 yd). The athlete jogs, runs, or sprints the distance between stations; stops at each station to perform a strength, flexibility, or muscular endurance exercise in a manner similar to that in actual circuit training; and continues on jogging, running, or sprinting to the next station. These courses are typically located in parks or in the country where there are many trees and hills. Such a training regimen can benefit almost any type of athlete and provide diversity to what might be an otherwise monotonous training regimen.

RESEARCH PERSPECTIVE 9.2

Tabata Training: The Original HIIT

Tabata, named for Dr. Izumi Tabata from Ritsumeikan University in Japan, is a high-intensity interval training (HIIT) protocol that incorporates brief, supramaximal (170% of $\dot{V}O_{2max}$) 20 s intervals followed by 10 s of rest into a 4 min exercise bout. In 1996, Dr. Tabata published the results of a study that found that—although just 4 min in duration—subjects who followed this HIIT protocol 4 days a week showed notable improvements in aerobic and anaerobic fitness, higher than those observed in subjects who performed classical endurance training (70% of $\dot{V}O_{2max}$) for 60 min a day.¹⁴ Subsequent research has shown that Tabata and other HIIT programs can be effectively used to increase aerobic and anaerobic fitness, promote fat loss, and improve health outcomes in a relatively short period of time.¹²

The HIIT model of brief bouts of aerobic conditioning at close to maximal intensity has become increasingly popular with competitive athletes to enhance both aerobic and anaerobic endurance while mimicking the swings in intensity that typically occur in competition. However, HIIT protocols can induce similarly large increases in fitness in recreationally active adults as well. Although the intensity of the intervals in the Tabata study was supramaximal, more recent studies have demonstrated that modified versions of the 20 s on/10 s off protocol can yield similar results with submaximal intensities that are still very high intensity, ranging from 80% to 95% of $\dot{V}O_{2max}$. These aerobic intervals are more appropriate for the general population and are frequently utilized in group fitness and personal training settings.

Although somewhat counterintuitive because of the high heart rates achieved, a growing number of fitness experts are suggesting that HIIT should be considered as a strategy to improve cardiovascular and metabolic health. Studies of nonathletes training with high-intensity intervals have documented improved metabolic rate and fat oxidation, decreased abdominal fat, greater insulin sensitivity, improved blood glucose, and reduced blood pressure after training. A report from a research team in the United Kingdom showed that doing 20 s Tabata-style intervals for just 3 min a week improved insulin sensitivity in young men.³ The authors reasoned that when insulin works more effectively, the muscles utilize more fatty acid oxidation for fuel, which in turn may result in greater utilization of fat even at rest. Indeed, the postexercise oxygen consumption following a 4 min bout of HIIT was double what it was just before exercise. The extra calories and potential for increased fat utilization resulting from HIIT may be an untapped resource to safely improve health outcomes, even in people who are not regularly physically active.

High-Intensity Interval Training (HIIT)

Traditionally, exercise physiologists have recommended one of three regimens to improve aerobic power: continuous exercise at a moderate to high intensity; long, slow (low-intensity) exercise; or interval training. However, a growing body of research suggests that **high-intensity interval training (HIIT)** is a time-efficient way to induce many adaptations normally associated with traditional endurance training. Scientists at McMaster University in Canada have studied the effects of training using short bursts of very intense cycling, interspersed with up to a few minutes of rest or low-intensity cycling for recovery.⁶ A common training mode employed is based on the Wingate test, a test that consists of 30 s of all-out cycling and generally produces mean power outputs that are two to three times higher than what subjects typically generate during a maximal oxygen uptake test.

A typical HIIT workout consists of four to six bouts of 30 s all-out cycling separated by a few minutes of recovery. Therefore, the total exercise time is as little as 2 min spread over a 20 min total time period. Several studies have now confirmed that performing six or so sessions of this type of interval training over a 2-week span can dramatically improve aerobic capacity in previously untrained individuals. The best feature of this type of training for busy exercisers is that such a regimen involves only 15 min total of all-out cycling within a total time commitment of 2.5 h!⁵

In addition to improving $\dot{V}O_{2\max}$, HIIT has been proven to have additional health benefits. Similar to continuous aerobic training programs, HIIT improves glucose control and insulin sensitivity, especially in individuals with (or at risk for) type 2 diabetes. HIIT has also been shown to improve vascular endothelial function, a measure of blood vessel health. In fact, studies have demonstrated that HIIT may be more effective than continuous, long-duration training in promoting metabolic and cardiovascular adaptations.¹⁰

HIIT for Athletes

Can highly fit individuals and endurance athletes also benefit from HIIT? In sedentary individuals, exercise training affects both the cardiovascular system and the muscles' oxidative enzyme capacity, resulting in an increase in $\dot{V}O_{2\max}$. In contrast, in already trained

individuals, an increase in exercise intensity close to or even slightly above $\dot{V}O_{2\max}$ is often necessary to elicit improvements in $\dot{V}O_{2\max}$ and performance.



There is growing evidence that inserting HIIT into an already high-volume traditional aerobic training program can further enhance performance.⁵ For example, when a group of well-trained cyclists replaced 15% of their normal training time with HIIT, they improved their peak power and speed during a 4 km time trial. Such improvement was seen after only six HIIT sessions inserted over a 4-week period.

RESEARCH PERSPECTIVE 9.3

Exploring the Mechanisms That Increase $\dot{V}O_{2\max}$ with HIIT

High-intensity interval training (HIIT) significantly improves maximal oxygen uptake ($\dot{V}O_{2\max}$). In untrained individuals, HIIT can increase $\dot{V}O_{2\max}$ to the same extent as moderate-intensity continuous training, despite the much shorter duration of exercise bouts. The mechanisms by which moderate-intensity aerobic training increases $\dot{V}O_{2\max}$ (e.g., greater blood volume, higher cardiac output, increases in stroke volume) are well characterized and discussed in detail in this chapter. However, despite the widely seen increases in $\dot{V}O_{2\max}$ in response to HIIT, the specific adaptations that underlie this

outcome have not been clarified. According to the Fick equation, increases in $\dot{V}O_{2\max}$ are mediated by increases in cardiac output or arterial–venous oxygen difference, or (a-v) O_2 difference. However, the scientific evidence is not clear whether HIIT improves one, the other, or both.

Recently, a study conducted by a team of scientists from Cal State San Marcos, SUNY Stony Brook, and the National College of Natural Medicine examined the cardiovascular adaptations to 6 weeks of HIIT in 71 healthy, active, young subjects.² The aims of this study were (1) how HIIT improves $\dot{V}O_{2\max}$ and (2) whether there is an optimal HIIT regimen that would produce the most benefit. In order to test this, the subjects were divided into four groups: a sprint interval-training group (SIT), a high-volume interval-training group (HIIT_{HI}), a periodized interval-training group (PER), and a control group that did not exercise (CON). For the first 10 exercise sessions, all subjects in the exercising groups performed the same HIIT protocol of 8 to 10 bouts of 60 s of cycling at 90% to 110% of peak power with 75 s of recovery between bouts. After this initial training, the subjects were randomized into their specific regimens for the remainder of the study. $\dot{V}O_{2\max}$, maximal cardiac output, stroke volume, heart rate, and the (a-v) O_2 difference were measured during progressive cycling exercise at the beginning of the study, at the midway point, and at the end of the study. Compared to the control group, all HIIT groups had a significant increase in $\dot{V}O_{2\max}$, and the magnitude of the increase in $\dot{V}O_{2\max}$ was not different between regimens. In all HIIT groups, maximal cardiac output and stroke volume increased, while maximal heart rate and the (a-v) O_2 difference did not change. Because HIIT increased maximal cardiac output but did not affect extraction, the study team concluded that HIIT increases $\dot{V}O_{2\max}$ by improving oxygen delivery through increased blood flow rather than by increasing the muscles' ability to extract oxygen.

Another study used a type of HIIT training called the *10-20-30 concept* (5 min intervals alternating between low speed for 30 s, moderate speed for 20 s, and close to maximal speed for 10 s) to assess whether 7 weeks of HIIT training could improve endurance performance, cardiovascular fitness, and overall physical health in a group of already well-trained individuals.⁷ The athletes who underwent the HIIT training increased their $\dot{V}O_{2\max}$ by 4% and increased their performance in both 1,500 m and 5 km runs, despite a ~50% reduction in their total training time. They also had a significant reduction in their total blood cholesterol, cholesterol fractions, and resting blood pressure. These results suggest that high-intensity interval training is capable of improving $\dot{V}O_{2\max}$, exercise

performance, and overall markers of cardiovascular health in previously trained individuals. This same group of researchers had previously demonstrated an increase in peripheral muscle membrane proteins and transporters and changes in muscle oxidative enzyme capacity in previously trained athletes who followed a traditional HIIT regimen.⁹

For busy athletes, HIIT training is easily achievable and effective at improving cardiovascular health as well as athletic performance. It may be also useful for athletes who wish to reduce their training time or volume before competition without sacrificing continued improvements in $\dot{V}O_{2\max}$ and performance.

Gibala and Jones recommend that for endurance athletes, 75% of total training volume be performed at continuous low intensities with 10% to 15% done using high-intensity intervals.⁵ While each bout of activity is anaerobic, the overall effect of HIIT is to stimulate adaptations similar to those with endurance training but in a shorter period of time and with less total work performed. Studies that have compared HIIT to a much higher total volume of traditional continuous endurance training have shown similar improvements in $\dot{V}O_{2\max}$ and cellular markers of improved aerobic capacity in untrained individuals. Adaptations were different in highly trained athletes. These adaptations are discussed in more detail in [chapter 11](#).

High-Intensity Interval Training in Team Sports

The science of training athletes depends heavily on the concept of specificity of training. However, designing a training regimen that provides sport-specific performance benefits while maintaining overall speed, fitness, and athletic skills—without overtraining—is often a difficult task. Adding HIIT to traditional endurance workouts has gained popularity for its benefits in improving sport-specific athletic performance,^{7,9} but most of the studies examining the effects of HIIT have been performed on athletes competing in individual sports, such as runners and cyclists, not in athletes who participate in team sports.

In order to determine whether HIIT training would be beneficial for performance, a group of elite soccer players were tested before, and again after, a 5-week HIIT intervention. The Danish Second Division players averaged 2.7 training sessions per week, with each session

lasting 3.6 h, and played one match per week. The HIIT, accomplished by carrying out drills without the ball, consisted of six to nine 30 s intervals per week at an intensity of 90% to 95% of $\dot{V}O_{2max}$. The number of HIIT intervals increased each week. The performance evaluation included a sprint test, an agility test, and repeated 20 m shuttle runs at progressively increasing speeds. After the HIIT intervention, the elite soccer players performed better on the shuttle runs by 11%, but performance on the sprint and agility tests was unchanged. Interestingly, after the HIIT interventions, the $\dot{V}O_{2max}$ was unchanged, but there was a reduction in the athletes' $\dot{V}O_2$ during running at a fixed speed of 10 km/h, indicating that running economy was improved by HIIT in these elite soccer players.



In Review

- Anaerobic and aerobic power training programs are designed to train the three metabolic energy systems: the ATP-PCr system, the anaerobic glycolytic system, and the oxidative system.
 - Interval training consists of repeated bouts of high- to moderate-intensity exercise interspersed with periods of rest or reduced-intensity exercise. For short intervals, the rate or pace of activity and the number of repetitions are usually high, and the recovery period is usually short. Just the opposite is the case for long intervals.
 - Both the exercise rate and the recovery rate can be closely monitored with use of a heart rate monitor.
 - Interval training is appropriate for all sports. The length and intensity of intervals can be adjusted based on the sport requirements.
 - Continuous training has no rest intervals and can vary from LSD training to high-intensity training. Long, slow distance training is very popular for general fitness training.
 - Fartlek training, or speed play, is an excellent activity for recovering from several days or more of intense training.
 - Interval-circuit training combines interval training and circuit training into one workout.
 - High-intensity interval training is a time-efficient way to induce many adaptations normally associated with traditional endurance training. In addition to consuming less time, it can be used to provide variety to the training.
 - High-intensity interval training has been shown to improve performance in already-trained individuals, including those participating in team sports such as soccer.
-

IN CLOSING

In this chapter, we reviewed general principles of training and the terminology used to describe these principles. We then learned the essential ingredients of successful resistance training and anaerobic and aerobic power training programs. With this background, we can now focus on how the body adapts to these different types of training programs. In the next chapter, we will see how the body responds to resistance training.

KEY TERMS

1-repetition maximum (1RM)
aerobic power
anaerobic power
continuous training
eccentric training
electrical stimulation
Fartlek training
free weights
high-intensity interval training (HIIT)
hypertrophy
interval-circuit training
interval training
isokinetic training
isometric training
long, slow distance (LSD) training
muscular endurance
plyometrics
power
principle of individuality
principle of periodization
principle of progressive overload
principle of reversibility
principle of specificity
principle of variation
static-contraction resistance training
strength
variable-resistance training

STUDY QUESTIONS

1. Define and differentiate the terms *strength*, *power*, and *muscular endurance*. How does each component relate to athletic performance?
2. Define aerobic and anaerobic power. How does each relate to athletic performance?
3. Describe and provide examples for the principles of individuality, specificity, reversibility, progressive overload, and variation.
4. What factors need to be considered when one is designing a resistance training program?
5. What would be the appropriate range for resistance and repetitions when one is designing a resistance training program targeted to develop strength? Muscular endurance? Muscular power? Hypertrophy?
6. Describe the various types of resistance training, and explain the advantages and disadvantages of each.
7. What type of training program would likely provide the greatest improvement for sprinters? Marathon runners? Football players?
8. What are some advantages of exercising in a group setting rather than alone?
9. Describe the various forms of interval and continuous training programs, and discuss the advantages and disadvantages of each. Indicate the sport or event most likely to benefit from each one.
10. High-intensity interval training has been shown to cause beneficial adaptations leading to improved performance. Describe those physiological adaptations.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

